

RESEARCH ARTICLE

Effective E-Paper Driving Waveform Design Based on Dynamic Programming

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Received: 27 April 2025 | **Revised:** 10 July 2025 | **Accepted:** 7 October 2025

Funding: This work was supported by the National Key Research and Development Program of China under grant 2022YFB3603704.

Keywords: electrophoretic display (EPD) | driving waveform | dynamic programming | response time

ABSTRACT

Electrophoretic display (EPD) driving involves controlling the electric field to drive the movement of charged particles in microcapsules, enabling image display and updates. The effectiveness of EPD technology relies on optimized driving waveform design to ensure display clarity and minimize response time. However, despite advancements, existing EPD techniques continue to face critical challenges such as ghost images, flicker, and long response times, all of which can detract from user experience. To address these issues, this paper proposes a novel EPD driving waveform design algorithm that integrates physical simulation and a dynamic programming approach. This algorithm is designed to identify the optimal driving path, thereby significantly reducing response time while effectively minimizing ghost images and flicker. Experimental results demonstrate that the proposed algorithm effectively mitigates ghost images and flicker, achieving shorter response time and higher display quality in 16-level grayscale print-style color EPD.

1 | Introduction

Electrophoretic display (EPD) technology offers several benefits, including low power consumption [1], wide viewing angles [2], sunlight readability [3], and high contrast [4], making it a popular choice for embedded devices [5], electric shelf labels [6], and so forth. However, its adoption is limited by issues such as long response time, ghost images, and flicker.

As illustrated in Figure 1, the structure of EPD operates on the principle of electrophoresis, where two colored nanoparticles are suspended in a non-polar solution, with microcapsules positioned between two electrode plates [7]. The movement of charged particles within these microcapsules is controlled by an electric field, enabling image display. By applying a specific

sequence of voltage pulses to the pixel electrode, the position of these charged particles can be manipulated, resulting in varying grayscale effects. This sequence of voltage pulses, known as the driving waveform, is crucial for controlling the electric field changes. A well-designed driving waveform can enhance response speed, reduce flicker, and optimize ghost images while maintaining image quality.

Traditional EPD waveforms follow three phases: image erasing, particle activation, and image writing [8]. When the voltage applied to the EPD is removed, the spacing of the particles and the viscosity of the solution change over time, leading to the formation of ghost images [9]. To mitigate ghost images, Kao et al. [10] merged the erase and activate phases and introduced response delay compensation, though at the cost of DC imbalance. Yang

Danling Kang and Xiaoyan Zhao have equally contributed to this work.

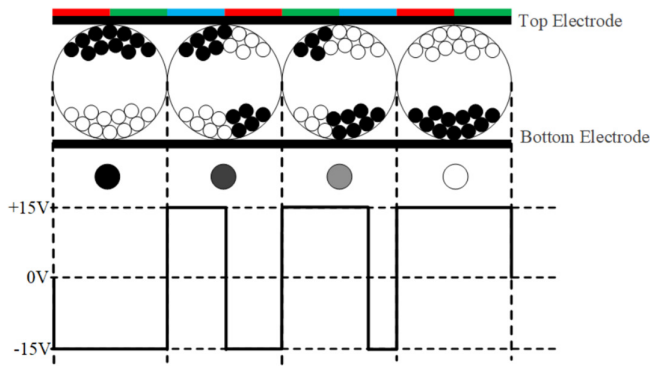


FIGURE 1 | Typical waveforms correspond to grayscale displays of EPD.

et al. [11] reduced ghost images by using a two-stage strong DC pulse driving waveform but did not shorten response time. Unstable driving waveforms can lead to flicker. To reduce flicker, Yi et al. [12] employed high-frequency square waves beyond human visual perception to suppress flicker, yet this approach did not optimize grayscale transitions. Li et al. [13] proposed an LTDR method to reduce flicker caused by refreshing. Duan et al. [14] proposed an adaptive generation method to create a driving waveform automatically which can reduce response time and visual flicker. Kao et al. [15] proposed a video halftone local refresh technology to alleviate flicker caused by global refresh but brought resolution decrease. Shorter response time requires optimizing voltage slope and transition speed. Kim et al. [16] introduced a multilevel voltage driving method to increase grayscale depth and reduce response time, but this added complexity. Bai et al. [17] and Wang et al. [18] optimized the erasing phase by selecting the shortest grayscale transition path, decreasing driving time. He et al. [19] introduced a strategy based on particle activation optimization to shorten response time.

While these approaches address specific limitations of EPD, they focus on isolated issues and lack a unified strategy to simultaneously optimize flicker, ghost images, and response time. This work proposes a comprehensive driving waveform strategy that integrates physical simulation and dynamic programming (DP) [20] to tackle these challenges. The main contributions of this work include the following:

1. A precise motion model of charged particles in EPD is developed using COMSOL Multiphysics simulations to quantify particle responses, providing a solid foundation for waveform optimization.
2. Leveraging physical modeling data and a DP algorithm, we realized phased optimization of the driving waveform by dynamically adjusting voltage and duration at each stage to achieve the optimal response path.

This approach overcomes the limitations of traditional designs that lack fine control in waveform optimization.

3. Experiments demonstrate that our algorithm delivers excellent performance: significantly reducing response time, effectively minimizing ghost images and flicker, providing outstanding display quality and refresh rates for 16-level grayscale color EPD.

2 | Driving Waveform Design

2.1 | Motivation

The design of EPD driving waveforms requires optimizing amplitude, pulse width, and duration to achieve high-quality display while minimizing response time, ghost images, and flicker. However, traditional methods often struggle to consider all these factors comprehensively. Higher voltages and longer durations are required for improved display quality, leading to increased power consumption despite a reduction in response time. Additionally, these methods often depend on manual tuning, resulting in inefficiencies and inconsistencies.

To address these challenges, we propose an approach that optimally adjusts voltage and duration at each phase using DP. The core principle of DP is to decompose a complex problem into simpler subproblems and obtain the global optimal solution by solving them sequentially. Using a bottom-up iterative approach, DP evaluates the value of each possible path at each stage, selects the optimal strategy accordingly, and ensures that the optimal solution at each stage contributes to the global optimum. The DP approach divides the original problem into smaller-dimensional subproblems by selecting at each step the decision set that returns the best outcome based on a given objective function [21]. In the context of EPD driving waveform, the DP algorithm enables simultaneous adjustments of both voltage and timing while maintaining display quality. This allows us to identify an optimal driving waveform that effectively balances image display quality and refresh rate. This method ultimately achieves comprehensive optimization of EPD display performance and efficiency.

2.2 | Data Collection

The key to DP design lies in the understanding of how charged particles behave under varying electric fields and how the dynamic influences the performance of EPD. This includes factors such as the time taken for particles to stabilize, the standard grayscale brightness of the EPD, and the refractive index of the device's surface.

We obtain the data through model simulations conducted in COMSOL Multiphysics [22]. To verify the reliability of the simulation results, we conduct an electronic ink experiment. First, a device based on electrophoresis principles is prepared, as depicted in Figure 2. We apply voltage to initiate the electronic ink display, achieving black-and-white image switching under the influence of the electric field, as shown in Figure 3. A colorimeter records real-time transformations of the electronic ink display, enabling us to collect vital data, such as response time. Next, we establish a simulation model of electronic ink in COMSOL Multiphysics, with boundary conditions defined in Table 1. The experimental model of electronic ink is depicted in Figure 4, in which (a) represents the simulated microcapsule shapes and (b) illustrates the detailed physics construction method used to mesh the model.

The model is subjected to the appropriate electric field to evaluate the consistency between simulation results and

experimental findings. We use the t -hypothesis test, as illustrated in Equation (1), to assess the validity of the simulation outputs, confirming that they fall within the 95% confidence interval. This verification reinforces the applicability of our simulations for practical design implementations:

$$t = (\bar{x} - u_0) / (s / \sqrt{n}). \tag{1}$$

Subsequently, the simulation model is utilized to collect key data to analyze particle and optical behaviors in the EPD. The transient analysis module applies different voltages to record the real-time motion data of the particles, and the particle tracing module exports the voltage-time curve, with termination conditions for the particle motion set to capture the time taken for the particles to reach a steady state. The ray tracing module then simulates the transmission behavior of light in the EPD, modeling light scattering and absorption across different grayscale states, and recording brightness data. Finally, the optical simulation module records the refractive index change curve in real-time while applying various voltage combinations. From this, the mechanism underlying these changes is extracted, and the amplitude of the refractive index change is calculated. Figure 5 illustrates the motion of particles in a COMSOL simulation.

Through the above processes, the following key parameter data-set for DP calculation is obtained:

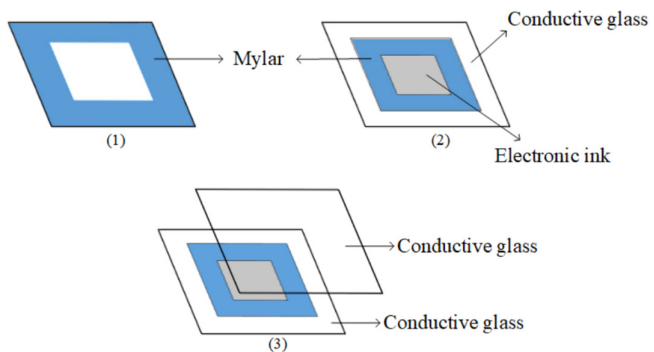


FIGURE 2 | Principle-based display device.



FIGURE 3 | Display of the electronic ink under voltage drive.

1. Duration:
The time required for particles to transition from the current grayscale level to the target grayscale level, reflecting the dynamic response speed of the display.
2. Brightness:
Brightness value of the EPD at various grayscale states, providing critical information regarding visual output under different driving conditions.
3. Refractive index:
The parameter quantifies the degree to which the EPD affects light propagation, including speed and direction of light, which directly impacts the optical performance of the display.

2.3 | Waveform Design Based on Dynamic Processing

The EPD driving waveform consists of three phases: erasing the previous image, activating charged particles, and writing the next image, as shown in Figure 6. During the image erasing phase (A to B), particles with different charge polarities move in a specific direction under the influence of the applied electric field. By applying a positive voltage, the charged particles transition to a black state, completely clearing the current content and providing a uniform initial state for the subsequent grayscale transitions. This phase is critical as it sets the groundwork for the clarity and fidelity of the images that follow. In the particle

TABLE 1 | Material settings.

Name	White particles	Black particles
Relative permittivity	2.486	2.487
Dynamic viscosity (mPas)	5.48	5.23
Density (g/mL)	0.83	0.82
Diameter (nm)	200	100

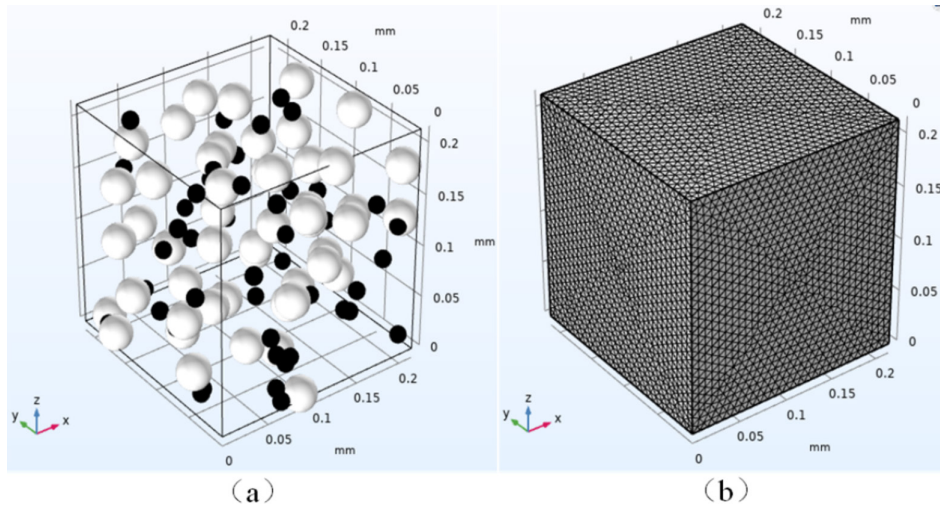


FIGURE 4 | Model build in COMSOL.

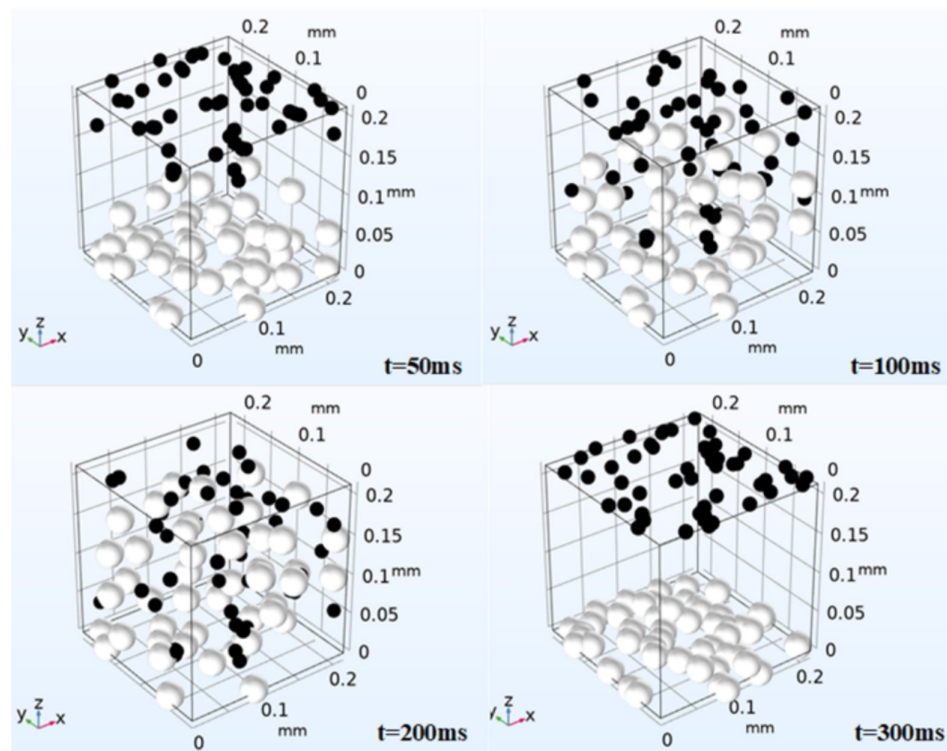


FIGURE 5 | The motion of simulated particles.

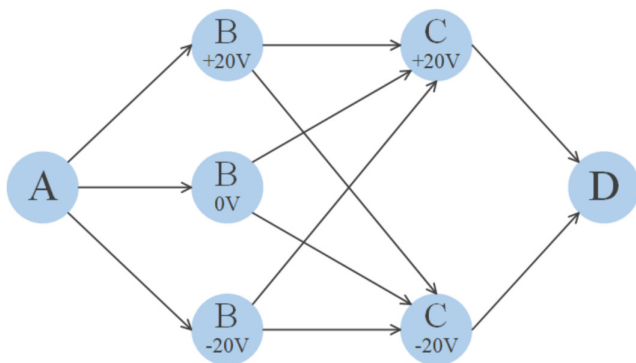


FIGURE 6 | Dynamic programming phases for EPD driving waveform design.

activation phase (*B to C*), a periodic electric field is created by alternating positive and negative voltages, which gradually drives the transition of particles from the initial grayscale to the target grayscale. This phase consists of multiple voltage cycles, with the exact number of cycles depending on the difference between the target and the current grayscale levels. Each cycle plays a crucial role in adjusting the energized particles to achieve the required visual output, emphasizing the importance of integrating precise voltage applications throughout this phase. Finally, the image writing phase (*C to D*) involves fine-tuning particle movement through the application of a stable voltage. This adjustment is essential to ensure that the particles settle into the desired target grayscale state accurately and without residual artifacts. The stability of the applied voltage during this phase is pivotal for achieving consistent grayscale rendering.

We abstract the 16 grayscales and three phases of the EPD into distinct states, define them as a quadruple, represented by the following:

$$dp[G][P] = \{V, t\}, \quad (2)$$

where $G \in \{G_0, G_1, \dots, G_{15}\}$, $P \in \{\text{erase}, \text{activate}, \text{write}\}$; to balance the response speed, power consumption limitation and hardware safety of EPD, we select the voltage $V \in \{+20V, -20V, 0V\}$, and t represents the duration of each applied voltage.

To identify the optimal path for the DP algorithm, the cost value of each path is evaluated to find the global path with the least total cost. The design of the EPD driving waveform thus requires the construction of a cost function that incorporates time efficiency, display quality, and user experience. By quantifying the varying costs associated with multiple voltage selections, we systematically guide the choice of the optimal voltage series. The driving time cost is represented by a duration parameter, denoted as T , which is directly linked to the refresh rate of the EPD. Display quality hinges on the phenomenon of residual charge accumulation. If the particles fail to reach their target positions completely, the charge distribution becomes uneven, resulting in brightness discrepancies. To quantify display quality, we employ luminance accuracy as a measure of cost. The standard luminance of a given target grayscale is denoted as $L_{\text{standard}}(G)$, while the current luminance achieved during the process is denoted as $L_{\text{current}}(G)$. The display quality cost is defined as follows:

$$Q(G) = w \cdot |L_{\text{current}}(G) - L_{\text{standard}}(G)|. \quad (3)$$

where w is the weight coefficient, which is used to adjust the influence of brightness interpolation on the overall cost function. User experience is primarily evaluated through visual comfort during the EPD refresh process. Flicker can occur as a result of abrupt changes in the refractive index, which are induced by significant fluctuations in charge distribution. As the driving process utilizes higher speeds, the resultant large voltage jumps lead to rapid movements of the particles, ultimately causing noticeable flicker. To quantitatively evaluate user experience U ,

we analyze the changes in the refractive index, and a more pronounced alteration correlates with a more severe flicker effect, thereby degrading user experience. In summary, the cost function can be expressed as follows:

$$f(V, T) = \lambda_1 T(V, t) + \lambda_2 Q(G) + \lambda_3 U(V, t), \quad (4)$$

where λ_1 , λ_2 , and λ_3 are weight parameters that adjust the importance of each factor in the decision-making process. Fast response is a crucial characteristic of EPD, which is why it has

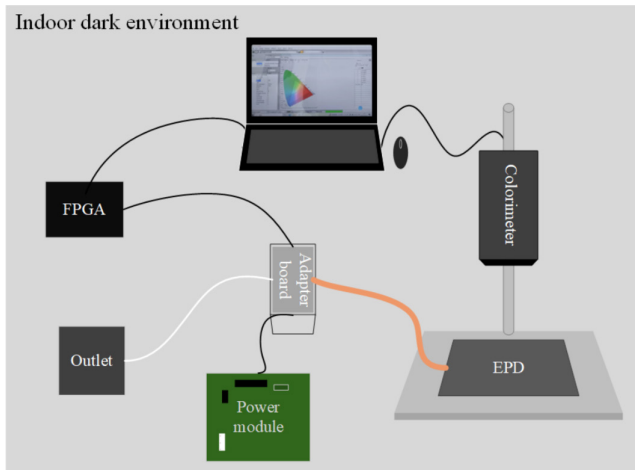


FIGURE 7 | Experimental platform.

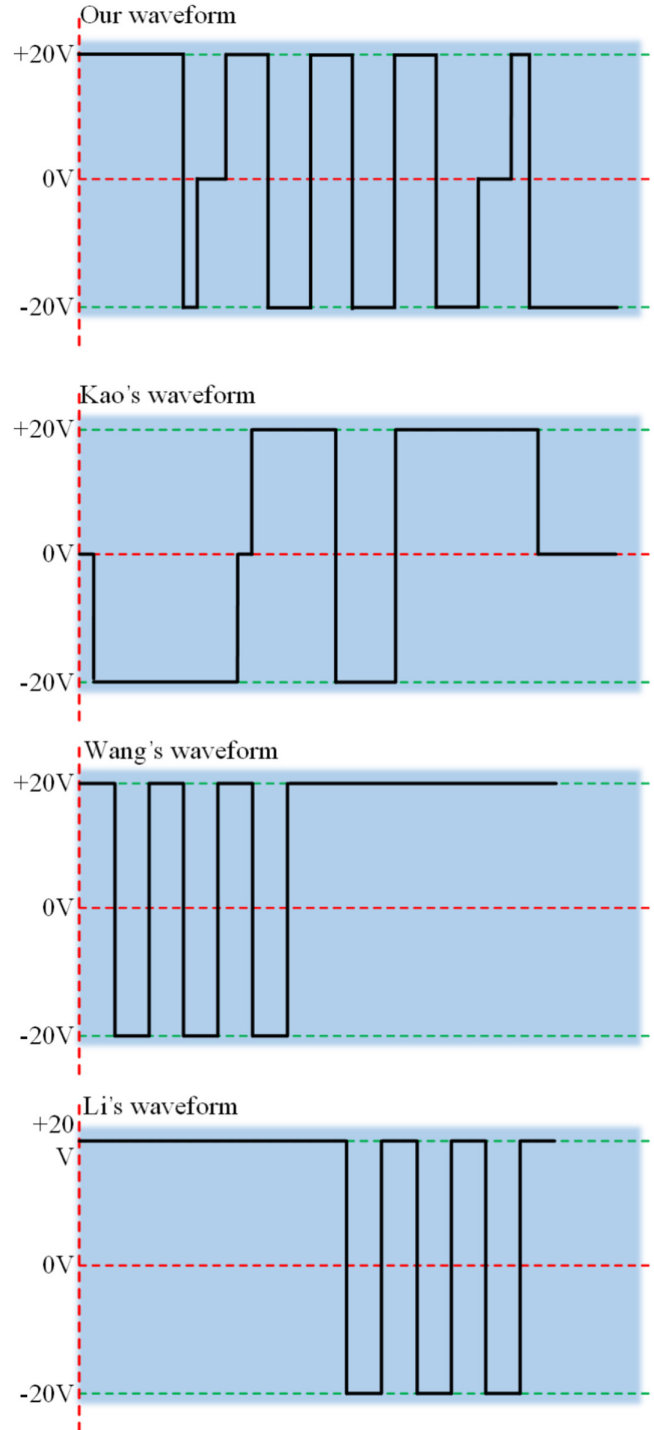


FIGURE 8 | Compared waveforms.

been assigned a weight of 0.6. Meanwhile, both display quality and user experience costs have been allocated a weight of 0.2 each. This weighting distribution is designed to optimize image display quality, minimize flicker, and effectively achieve the fast response objective.

Once the definitions are established, DP calculations are performed according to the following steps:

1. Initial state and boundary condition setting:

$$dp[G_0][erase] = 0. \quad (5)$$

The image erasing phase begins at grayscale G_0 , allowing us to define the initial state as:

After the image erasing phase, particles gradually transition from G_0 to the target grayscale G_i , which is subsequently followed by the particle activation and image writing phases:

$$dp[G_i][write] = \min(dp[G_i][P]). \quad (6)$$

TABLE 2 | Response time for each waveform.

Waveform	Response time/ms
Kao's waveform [24]	107.022
Wang's waveform [18]	103.514
Li's waveform [13]	91.360
Ours	90.853

2. State transition equation modeling:

The state transition influences the path selection from the current state to the next. The path cost is calculated according to the cost function linked to each phase, enabling the identification of the optimal transition methods. Each transition is dictated by voltage selection, voltage duration, and the path generation value, which is defined as follows:

$$dp[G_i][P] = \min_{V,T} (dp[G_{i-1}][P] + f(V, T)), \quad (7)$$

where $dp[G_i][P]$ is the minimum cost of the current grayscale G_i in phase P , $dp[G_{i-1}][P]$ is the minimum cost from the previous phase, and $f(V, T)$ is the cost function. During state transitions, the algorithm monitors particle positions and residual charge distributions. The charge state is indirectly represented in the cost function through variations in the refractive index and brightness. This approach ensures that the algorithm actively manages charge accumulation throughout the optimization process.

3. Optimal path backtracking:

The backtracking process, initiated from the final state back to the initial state, allows us to identify the optimal decision sequence. This process meticulously preserves a record of the optimal choice made at each phase, allowing the optimal path to be traced stepwise according to the recursive formulation:

$$Opt[P] = \arg \min_P (dp[G_i][P]), \quad (8)$$

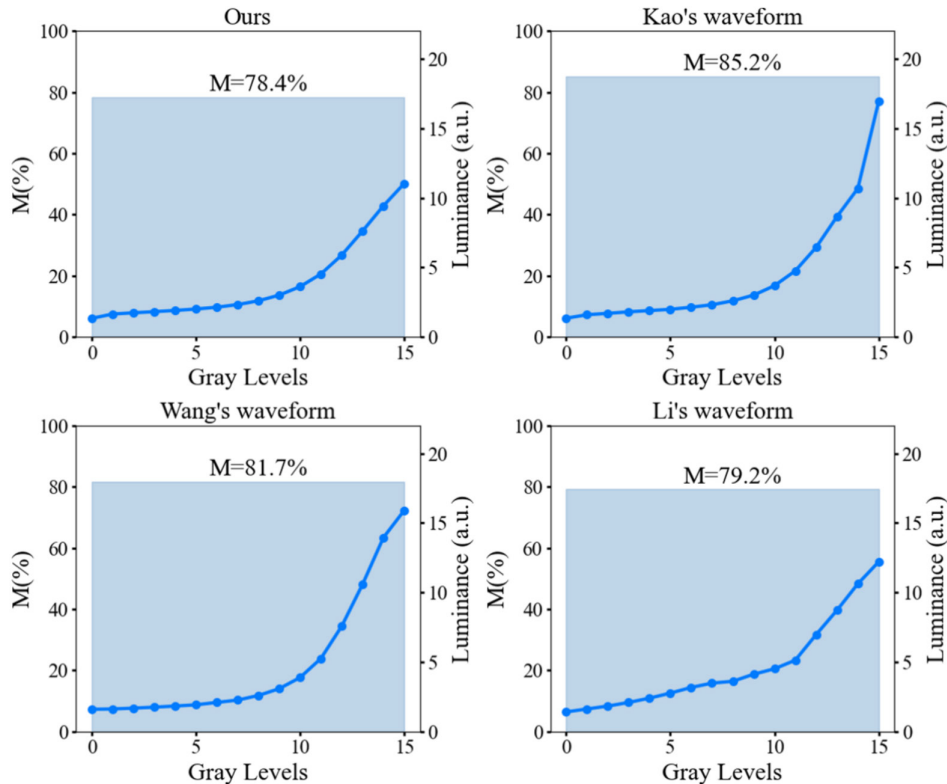


FIGURE 9 | Experiment results of flicker.

where $Opt[P]$ denotes the optimal path selection and $dp[G_i][P]$ denotes the target grayscale G_i at the lowest cost in the final phase P . Through continuous backtracking, we systematically capture the optimal choices for each phase, ultimately resulting in the global optimal voltage sequence.

3 | Experimental Results

The performance of printed-color EPD is determined by response time, flicker, and ghost images. To evaluate the effectiveness of our driving waveform, we construct an experimental platform, as depicted in Figure 7. To eliminate the reflection on the EPD, we create an indoor dark environment and use two table lamps to provide indirect lighting, ensuring that the light does not directly strike the EPD. Additionally, since the system cannot conduct experiments in direct sunlight, we employ strong artificial lighting to simulate sunlight, allowing us to effectively demonstrate the display performance of the EPD under those conditions. This platform comprises a computer (Dell, China), a colorimeter (Ademesy), and a driver board. The EPD used is EC133UJ1, featuring a resolution of 1600×1200 and a refresh rate of 75 Hz.

During testing, the driving waveform is prepared by waveform editing software and subsequently imported into the FPGA main control module through a USB interface. The EPD is driven through an adapter board, and the colorimeter collects real-time data. These data are then transmitted to the computer for analysis using Iliad software. To assess the effectiveness of the proposed driving waveform, we conduct comparative tests against three compared waveforms to drive the EPD, as illustrated in Figure 8.

In order to test the response time of different waveforms during the grayscale conversion process, we apply these waveforms to refresh the EPD to extreme black and white. Simultaneously, the colorimeter records the time taken for each transition, specifically, the time t_1 for the EPD to transition from black to white and t_2 for the transition from white to black. The average response time is calculated as $t = (t_1 + t_2) / 2$ [23]. Multiple measurements for each waveform were conducted to ensure reliability, resulting in the average response time data summarized in Table 2.

In addition, we quantitatively evaluate the visual flicker intensity using the light source modulation depth [13]. The light



FIGURE 10 | Experiment results of ghost images.

source modulation depth, also known as the flicker percentage, serves as both a direct and indirect measure of flicker intensity exhibited on the screen. It is calculated in the following:

$$M = ((I_{max} - I_{min}) / (I_{max} + I_{min})) \times 100\%, \quad (9)$$

where I_{max} and I_{min} represent the maximum and minimum brightness value of the EPD during grayscale conversion, respectively. A larger M indicates a more pronounced brightness change, leading to higher flicker intensity. We record the reflected luminance data from the EPD when different waveforms drive the transition from G_0 to G_{15} , as presented in Figure 9.

Finally, we carry out ghost images test to further evaluate the influence of different waveforms on the display quality of EPD. Each waveform is employed to sequentially display a series of images without refreshing between images, which allows us to observe any ghost images. The results of the experiment are shown in Figure 10.

Through the experiments described above, Kao's waveform adopts an extreme driving strategy that effectively achieves complete erasure of the image, and significantly alleviates the ghost images. However, this method induces substantial particle migration, leading to the longest response time among the tested waveforms. Wang's waveform, with an M -value of only 81.7%, improves flicker of the EPD but shows minimal effect on ghost images, making it the least effective among the four waveforms while providing limited improvement in response time. Li's waveforms greatly reduce response time and significantly alleviate ghost images while exhibiting relatively low flicker intensity. Nevertheless, the waveform results in noticeable image graying and reduced overall image contrast. The driving waveform proposed in this paper demonstrates a slightly lower M -value but achieves the most significant improvement in flicker reduction and exhibits the shortest response time. Regarding ghost images, our waveform effectively minimizes ghost images without compromising the original image display quality, thus balancing display quality, user experience, and response speed.

4 | Conclusion

In this letter, we propose a novel algorithm for designing EPD driving waveforms to enhance display performance. By constructing a simulation model, we accurately predict particle motion, gather necessary data, and apply DP for multistage processing, optimizing each waveform stage to achieve the globally optimal waveform sequence. The integration of physical simulation techniques with DP enables our algorithm to effectively address the critical challenges commonly associated with EPDs, such as long response time, ghost images, and flicker, which significantly impact user experience and overall display quality. Comprehensive experimental studies clearly indicate that our newly designed waveform significantly surpasses other comparative methods. It effectively reduces the incidence of ghost images and flicker while achieving a notably quicker response time, thereby improving both display performance and user satisfaction.

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